

Where Does the Union Bound Go?

Best-Arm Identification and Strong FWER Control

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Abstract

In fixed-confidence best-arm identification, proofs often use a union bound across the competing arms. From a multiple-testing point of view this can look puzzling: if the best arm is unique, only one hypothesis of the form “arm i is best” can be true. Why then should there be a Bonferroni-type factor of $K - 1$? The answer is that there are two natural ways to orient the hypotheses. In one orientation, best-arm identification is literally a strong familywise-error-rate (FWER) problem with $K - 1$ true nulls. In the opposite orientation, exactly one null is true, but a pairwise implementation can falsely reject that one null through any of $K - 1$ comparisons. Thus the multiplicity has not disappeared; it just pops up in different places. This note makes the equivalence explicit in the terminology of both communities.

1 The puzzle

Suppose there are K arms with means

$$\mu_1, \dots, \mu_K.$$

An “arm” is simply one candidate option (for example, a treatment or system) from which the algorithm can collect observations. Let

$$\Theta_{\text{uniq}} := \left\{ \mu : \left| \arg \max_{i \in \{1, \dots, K\}} \mu_i \right| = 1 \right\}$$

be the unique-best-arm parameter space, and assume throughout that $\mu \in \Theta_{\text{uniq}}$. Then the unique best arm is

$$i^* = \arg \max_{i \in \{1, \dots, K\}} \mu_i.$$

In fixed-confidence best-arm identification (BAI), an adaptive algorithm samples the arms, stops at a data-dependent time τ , and returns an arm $\hat{i}_\tau$. We use δ -correctness below for the error-control requirement

$$\Pr_\mu(\hat{i}_\tau \neq i^*) \leq \delta \quad \text{for every instance with a unique best arm.} \quad (1)$$

In the standard fixed-confidence BAI definition, a δ -PAC strategy is additionally required to stop almost surely, i.e. $\Pr_\mu(\tau < \infty) = 1$; PAC stands for “probably approximately correct”. See, for example, [Garivier and Kaufmann \(2016\)](#).

A standard BAI proof often reaches an event of the form

$$\{\hat{i}_\tau \neq i^*\} \subseteq \bigcup_{j \neq i^*} \mathcal{E}_j$$

and then uses

$$\Pr_{\mu}(\hat{i}_{\tau} \neq i^*) \leq \sum_{j \neq i^*} \Pr_{\mu}(\mathcal{E}_j).$$

This is simply the union bound, and it commonly leads to an error allocation such as $\delta/(K-1)$ across the competing events; in logarithmic stopping thresholds this typically appears as an additive $\log(K-1)$ contribution. Modern analyses can make the underlying time-uniform concentration much sharper, but a union over the $K-1$ wrong answers still appears in standard BAI calibrations (Kaufmann and Koolen, 2021).

Now compare this with a familiar multiple-testing observation. If a family of null hypotheses is logically constrained so that at most m_0 of them can be true simultaneously, a Bonferroni proof of strong familywise-error control only needs to sum over those m_0 true nulls. But among the statements

“arm 1 is best”, $\dots$, “arm K is best”,

exactly one is true. So why is the BAI proof paying for $K-1$ events?

The apparent contradiction comes from implicitly switching between two different families of null hypotheses.

2 Multiple testing: strong FWER control

For a family of null hypotheses $H_1, \dots, H_m$, let $\mathcal{R}$ be the set of rejected hypotheses and define the index set of true nulls by $\mathcal{H}_0(\mu) := \{i : \mu \in H_i\}$. Strong familywise-error-rate (FWER) control at level α means

$$\Pr_{\mu}(\mathcal{R} \cap \mathcal{H}_0(\mu) \neq \emptyset) \leq \alpha \quad \text{for every } \mu. \quad (2)$$

In words: the probability of rejecting at least one true null is at most α , whatever the configuration of true and false nulls.

The elementary Bonferroni argument is

$$\Pr_{\mu}(\mathcal{R} \cap \mathcal{H}_0(\mu) \neq \emptyset) \leq \sum_{i \in \mathcal{H}_0(\mu)} \Pr_{\mu}(H_i \text{ is rejected}).$$

The sum is over the *true* nulls, not automatically over every hypothesis in the family. Logical information about which nulls can be true together can therefore matter substantially; this is one of the motivations behind the partitioning principle and related structured multiple-testing methods (Finner and Strassburger, 2002).

The same basic idea—use the logical restrictions among the hypotheses rather than pretending that every truth configuration is possible—also underlies classical improvements of Bonferroni such as Shaffer’s procedure and is closely related to closed testing (Shaffer, 1986; Marcus et al., 1976).

The partitioning principle

Suppose the parameter space is split into disjoint regions $P_1, \dots, P_M$ that cover all possibilities. Exactly one region is true. If every P_m is tested at level α , there is no need to divide α by M : the event “reject at least one true partition hypothesis” is simply the event “reject the unique true region”, which has probability at most α . Now suppose an original null hypothesis H is the union of several of these regions. Reject H only when *all* partition regions contained in H have been rejected. If H is true, the unique true cell lies inside H , so rejecting H necessarily entails rejecting that true cell. Hence the induced decisions on the original hypotheses strongly control FWER at level α . This is the partitioning principle: first encode the logical structure into mutually exclusive cases, and only then test (Finner and Strassburger, 2002).

3 Orientation 1: “arm i is not best”

Define

$$H_i^- := \{\mu \in \Theta_{\text{uniq}} : i \neq i^*(\mu)\}. \quad (3)$$

The null H_i^- says that arm i is *not* the best arm. If a BAI procedure recommends arm i , interpret this as rejecting H_i^- . Since the procedure returns one arm, its rejection set is simply

$$\mathcal{R} = \{\hat{i}_\tau\}.$$

Suppose the true best arm is i^* . Then

$$H_{i^*}^- \text{ is false, } \quad H_i^- \text{ is true for every } i \neq i^*.$$

So there are exactly $K - 1$ true nulls.

Fact 1. *For the family (3), the BAI error event is exactly the FWER event:*

$$\{\hat{i}_\tau \neq i^*\} = \{\mathcal{R} \cap \mathcal{H}_0(\mu) \neq \emptyset\}.$$

Consequently, δ -correct BAI is strong FWER control at level δ for this logically constrained family over the parameter space Θ_{uniq} .

In this orientation there is therefore no puzzle at all. The multiple-testing rule “only pay for hypotheses that can be true at the same time” says that a naive Bonferroni proof may have to pay for $K - 1$ hypotheses, because $K - 1$ of these nulls *are* simultaneously true.

4 Orientation 2: “arm i is best”

Now reverse the hypotheses. Define

$$G_i := \{\mu \in \Theta_{\text{uniq}} : i = i^*(\mu)\}. \quad (4)$$

The G_i form a partition of the unique-best-arm parameter space. Exactly one of them is true. Thus the G_i are literally partition hypotheses in the sense of the box in Section 2.

Imagine an elimination algorithm: reject G_i when there is sufficient evidence that arm i cannot be best, eliminate arm i , and stop when only one arm remains. In this formulation, the multiple-testing intuition is correct in a very literal way.

Fact 2. *If each G_i is tested by an anytime-valid level- δ test, meaning that its rejection event satisfies*

$$\sup_{\mu \in G_i} \Pr_{\mu}(\exists t : G_i \text{ is rejected by time } t) \leq \delta,$$

with respect to the filtration generated by the adaptively sampled bandit experiment, then an elimination procedure that returns the last non-rejected cell is δ -correct. No multiplicity correction across $G_1, \dots, G_K$ is needed.

So where did the $K - 1$ go? It appears when we ask how a test of the *one* true composite hypothesis G_{i^*} is constructed.

Equivalently,

$$G_i = \bigcap_{j \neq i} \{\mu_i > \mu_j\}. \quad (5)$$

Thus “arm i is best” is an intersection of $K - 1$ pairwise statements. A pairwise elimination rule rejects G_i as soon as one competitor has produced sufficient evidence against it. If $R_{j,i}$ denotes the event that the pairwise comparison with j eliminates i , then

$$\{G_i \text{ is rejected}\} = \bigcup_{j \neq i} R_{j,i}$$

for such a construction. Under the true best-arm null G_{i^*} , all $K - 1$ pairwise inequalities

$$\mu_{i^*} > \mu_j, \quad j \neq i^*,$$

are true. Hence

$$\Pr_{\mu}(G_{i^*} \text{ is rejected}) \leq \sum_{j \neq i^*} \Pr_{\mu}(R_{j,i^*}). \quad (6)$$

The Bonferroni correction has moved inside the test of the single true composite null.

5 What this says and what not, and related work

The point of the observation is logical, not that every union bound used in BAI is necessary or sharp. Once we recognize that “arm i is best” is one composite null, it is natural to ask whether it can be tested directly rather than by $K - 1$ separately calibrated pairwise tests. Two classical statistical traditions are especially close to this question: ranking and selection, and multiple comparisons with the best (MCB). Bechhofer’s early ranking-and-selection formulation explicitly recast ranking normal means as a multiple-decision problem (Bechhofer, 1954). Paulson subsequently gave a sequential procedure for selecting the population with the largest mean in which inferior populations can be eliminated as sampling proceeds (Paulson, 1964). The MCB literature studies comparisons of each treatment with its competitors and includes both sequential and simultaneous procedures (Hsu and Edwards, 1983; Edwards and Hsu, 1983; Hsu, 1984). Hsu’s stepwise MCB procedure is particularly close in spirit to an elimination viewpoint: it successively identifies treatments as strictly not the best (Hsu, 1992). These procedures share the same geometry as BAI: statements about a treatment being best are built from comparisons with its competitors. On the multiple-testing side, the partitioning and closed-testing viewpoints make explicit that logical relations among hypotheses can reduce multiplicity, while Shaffer-type procedures exploit the restricted sets of hypotheses that can be true simultaneously (Marcus et al., 1976; Shaffer, 1986; Finner and Strassburger, 2002).

The factor $K - 1$ can be seen particularly cleanly in the development of fixed-confidence BAI. In the Bernoulli case, Theorem 10 of Garivier and Kaufmann (2016) proves δ -correctness of Chernoff’s stopping rule with threshold

$$\beta(t, \delta) = \log\left(\frac{2t(K - 1)}{\delta}\right).$$

The proof first fixes the true best arm and then takes a union over the $K - 1$ possible wrong winners. The $K - 1$ there counts the $K - 1$ distinct ways in which the one true best-arm statement can be contradicted by a competitor.

Kaufmann and Koolen (2021) sharpen this analysis substantially. Their mixture-martingale inequalities are time-uniform and can combine evidence from several arms, avoiding a cruder arm-by-arm concentration argument. Nevertheless, in their Section 5.1 the BAI error event, with arm 1 taken to be the true best arm, is again bounded by an event of the form

$$\{\exists t, \exists a \neq 1 : \text{the pair } \{1, a\} \text{ crosses its boundary}\}.$$

They then explicitly apply a weighted union bound over the $K - 1$ subsets $\{1, 2\}, \dots, \{1, K\}$. More precisely, their resulting threshold is

$$\hat{c}_t(\delta) = 6 \log\left(\log\left(\frac{t}{2}\right) + 1\right) + 2C_{\exp}\left(\frac{1}{2} \log\left(\frac{K - 1}{\delta}\right)\right),$$

where $C_{\exp}(x) \sim x$ as $x \rightarrow \infty$. Thus the multiplicity enters through $\log((K - 1)/\delta)$ as the argument of the calibration function. Its dependence on time is much sharper than the earlier

$\log(2t(K-1)/\delta)$ boundary. This is exactly the distinction highlighted in Section 4: the concentration within a pair can be sharpened without making the $K-1$ possible pairwise witnesses disappear.

The same paper introduces the *rank* of a pure-exploration identification problem (that is, a sequential learning problem whose aim is to identify a correct answer rather than to maximize cumulative reward). BAI has rank 2, because to show that a proposed arm i is not best it suffices to exhibit one competitor a with $\mu_a > \mu_i$; each individual witness therefore involves only two arms. But there may still be $K-1$ such witnesses. The rank and the number of possible witnesses are consequently different notions: rank 2 explains why pairwise statistics suffice, while the union over the possible competitors explains why $K-1$ can remain in a nonasymptotic error calibration (Kaufmann and Koolen, 2021).

This also clarifies what should *not* be concluded. These proofs do not say that every BAI method must incur a multiplicative $K-1$ loss in sample complexity. For fixed K and $\delta \downarrow 0$, the classical BAI lower bound and the asymptotically optimal Track-and-Stop analysis are governed at leading order by $\log(1/\delta)$; inserting $K-1$ inside the logarithm changes only a lower-order term (Garivier and Kaufmann, 2016). Also, Bonferroni is not the only possible way, as the vast multiple testing literature shows, to construct a test of G_i : a joint test can exploit dependence and the geometry of the full composite null. There are concrete examples of this possibility in related pure-exploration problems: Katz-Samuels et al. (2020) use empirical-process tools and the Gaussian width of the arm set to obtain linear and combinatorial bandit guarantees that avoid an explicit union bound over the number of arms. This is not the same as removing the $K-1$ term from ordinary unstructured BAI, but it illustrates that joint geometric control can replace an arm-by-arm union bound in a closely related identification problem.

Finally, there are nearby BAI formulations that make the multiple testing language even more explicit. A representative confidence-bound method is LUCB, whose stopping logic compares lower and upper confidence bounds (Kalyanakrishnan et al., 2012), while Garivier and Kaufmann (2021) study sequential identification directly as testing among possibly overlapping hypotheses. So the similarities are plenty: both communities are organizing error across logically related candidate answers, but they often place the multiplicity at different levels of the construction.

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